Congratulations! You passed!

Grade received 100% **Latest Submission Grade** 100% **To pass** 80% or higher

Go to next item

Generalized Additive Models: Inference and Data Analysis

Review Learning Objectives

Submit your assignment

Due Jun 16, 11:59 PM IST

Try again

Receive grade

To Pass 80% or higher

Your grade 100%

View Feedback

We keep your highest score

Like Dislike Report an issue

1. In a generalized additive model, if a smooth term has an effective degrees of freedom close to 1, then that term should enter linearly into the model. **3 / 3 points**

- (•) True
- ( ) False

Correct

2. The tests in the summary of the gam() function (in the mgcv package) associated with the smooth terms are (approximately) F-tests that test the hypothesis that the given smooth term is zero vs nonzero. **3 / 3 points**

- (•) True
- ( ) False

Correct

3. The adjusted R^2 is reported as a percentage, and is equal to $R^2 = 1 - \frac{RD}{ND}$, where RD is the residual deviance and ND is the null deviance. **3 / 3 points**

- ( ) True
- (•) False

Correct

4. In the context of generalized additive models, the adjusted R^2 is reasonable to use for model comparisons. **3 / 3 points**

- (•) True
- ( ) False

Correct

5. The trees data frame has 31 observations on 3 variables. **3 / 3 points**

1. **Girth**: Tree diameter in inches
2. **Height**: Height in ft
3. **Volume**: Volume of timber in cubic ft

Consider a GAM fit to the data, where Volume is the response and Girth and Height are predictors.

```
Family: Gamma
Link function: log

Formula:
Volume ~ s(Height) + s(Girth)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.27570    0.01492   219.6   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
            edf Ref.df      F  p-value
s(Height) 1.000  1.000  31.32 3.92e-06 ***
s(Girth)  2.422  3.044 219.28  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.973   Deviance explained = 97.8%
GCV = 0.0080824  Scale est. = 0.006899  n = 31
```

Height should enter the model parametrically.

- (•) True
- ( ) False

Correct

6. The trees data frame has 31 observations on 3 variables. **3 / 3 points**

1. **Girth**: Tree diameter in inches
2. **Height**: Height in ft
3. **Volume**: Volume of timber in cubic ft

Consider a GAM fit to the data, where Volume is the response and Girth and Height are predictors.

```
Family: Gamma
Link function: log

Formula:
Volume ~ s(Height) + s(Girth)

Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.27570    0.01492   219.6   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
            edf Ref.df      F  p-value
s(Height) 1.000  1.000  31.32 3.92e-06 ***
s(Girth)  2.422  3.044 219.28  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.973   Deviance explained = 97.8%
GCV = 0.0080824  Scale est. = 0.006899  n = 31
```

The small p-value associated with Girth suggests that it should enter the model nonparametrically.

- ( ) True
- (•) False

Correct